



Chhattisgarh Swami Vivekanand Technical University

Library Management System



ANIL NAI .
300012824009 .
CSE(DS) .

◆ Project Title

Library Management System using C++

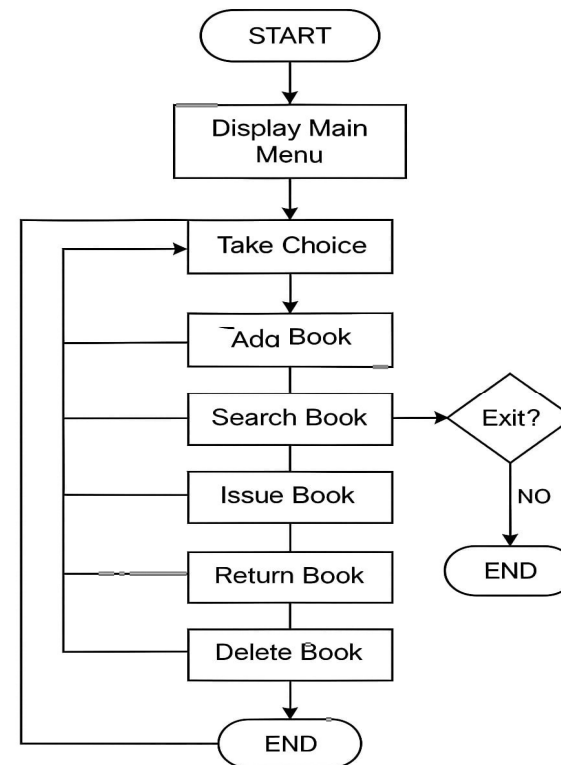
◆ Objective of the Project

The main objective of this project is to **manage library book records efficiently.**

It helps in performing basic library operations such as:

1. Adding new books
2. Displaying all books
3. Searching a book by ID
4. Issuing a book
5. Returning a book
6. Deleting a book record

All data is stored permanently using **binary file handling.**

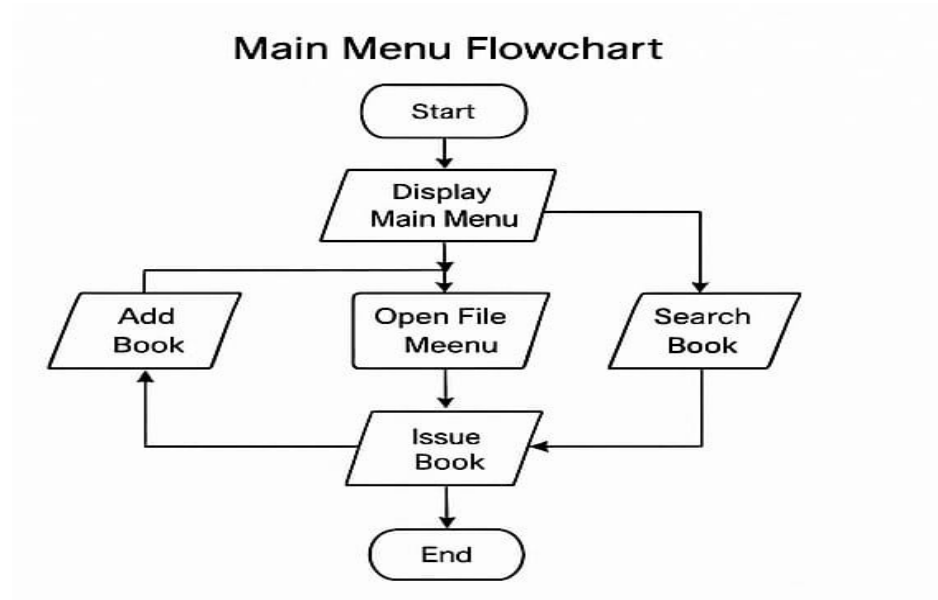


◆ Main Function (main())

This function controls the complete program.

Menu Options:

1. Add Book
2. Display Books
3. Search Book
4. Issue Book
5. Return Book
6. Delete Book
7. Exit



The program runs continuously using a while True loop until the user chooses **Exit**.

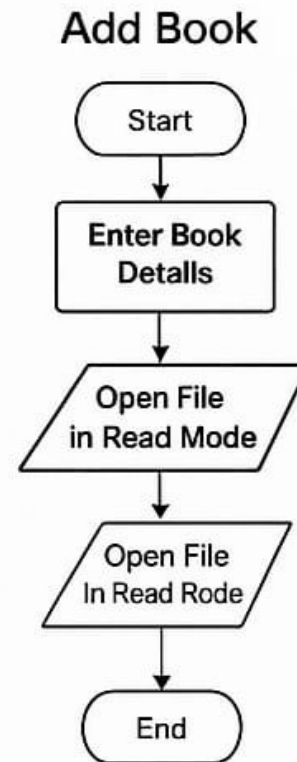
1 Add_book()

Purpose: Add a new book to the library

Working:

1. Takes book details from the user
2. Creates a Book object
3. Stores it in books.dat using pickle

➡ **File Mode:** "ab" (append binary)



2 Display_books()

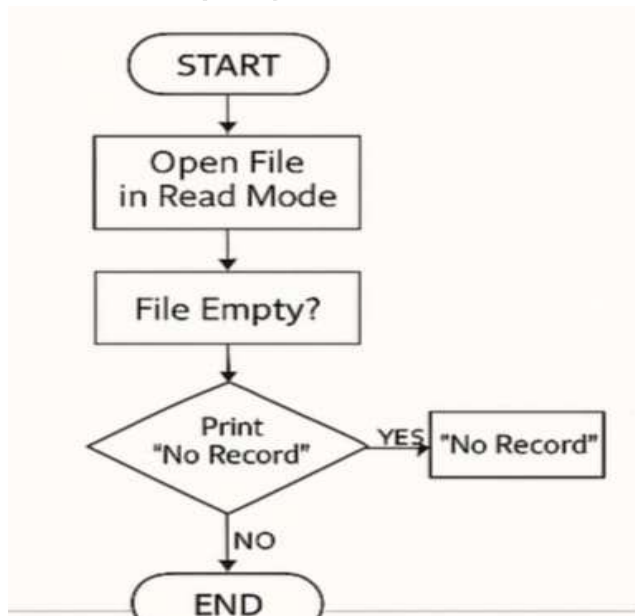
Purpose: Display all available books

Working:

1. Checks if the file exists
2. Reads all book objects one by one
3. Displays each book using the display() method

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Display books



3 Search_book()

Purpose: Search a book by Book ID

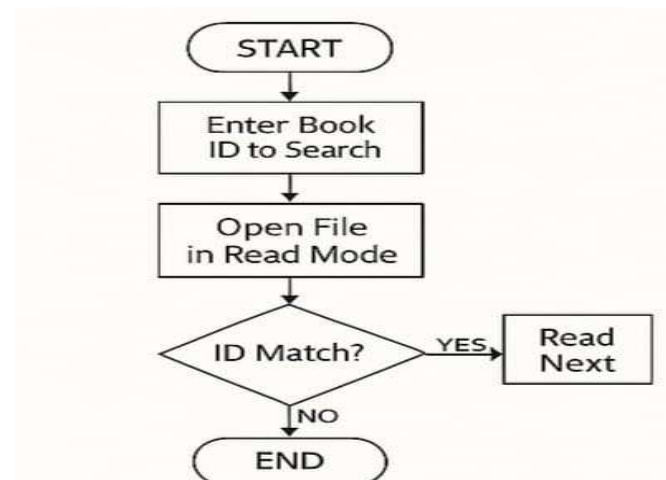
Working:

1. Takes Book ID from the user
2. Searches the file record by record
3. Displays the book if found

→ If not found, it prints **“Book not found!”**

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Search book



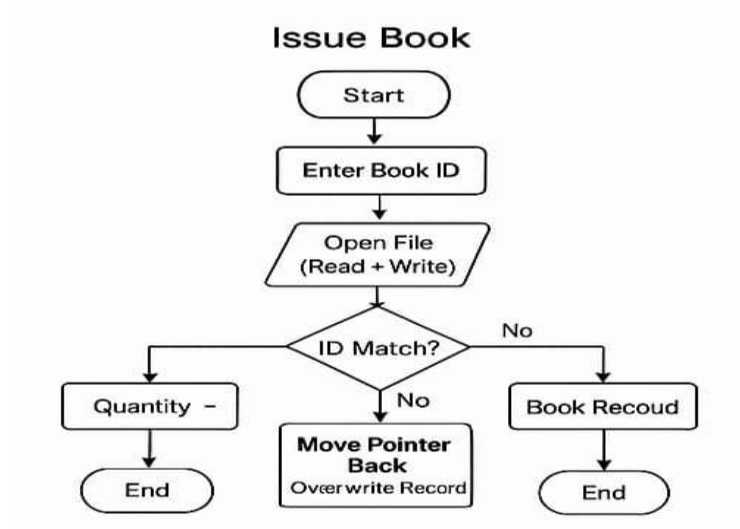
4 Issue_book()

Purpose:

To issue a book from the library.

Working:

1. Takes Book ID from the user
 2. Reads all book records from the file
 3. Matches the entered Book ID
 4. If quantity > 0 , decreases quantity by 1 and issues the book
- If quantity is 0, it prints **“Book out of stock!”**
 - If book is not found, it prints **“Book not found!”**

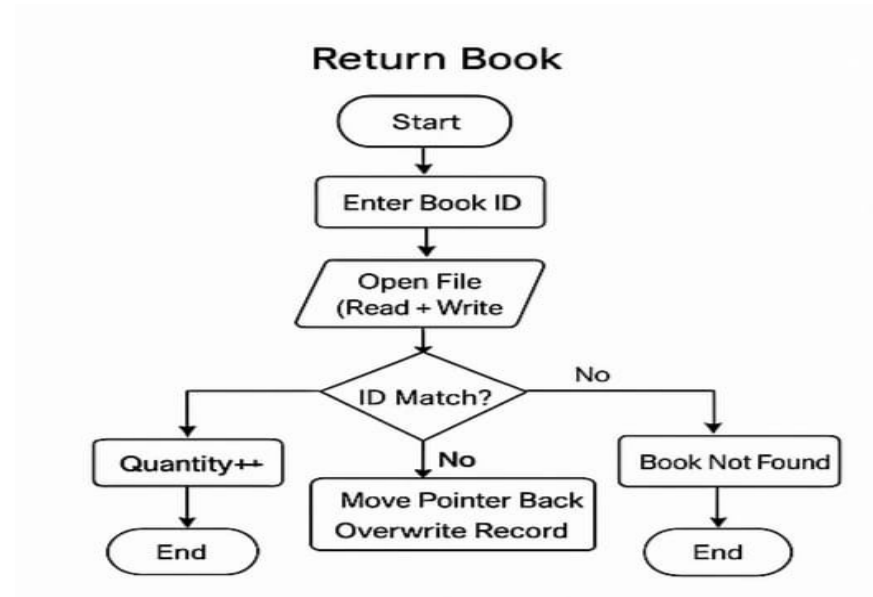


5 Return_book()

Purpose: Return a book

Working:

1. Takes Book ID
2. Increases the quantity by 1
3. Updates the file with new quantity



6 Delete_book()

Purpose: Delete a book record

Working:

1. Reads all books from the file
2. Removes the book with the given ID
3. Writes remaining books back to the file

